

The Influence of Intonation Changes Based on Heart Rate on Empathy Formation with a Chatbot

Ritsumeikan University, Kusatsu, Shiga, Japan

Yuta Noguchi, Makoto Nokata

Background

Having friendly conversations with robots and machines

➔ Difficulty in **sympathizing** with people

Know the others , Adapt to the others

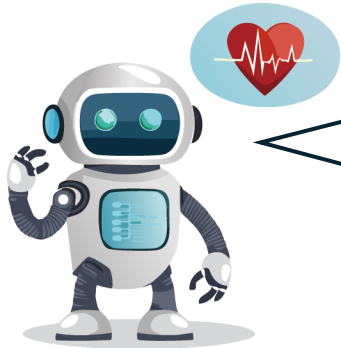
How to know

Human : Using five senses

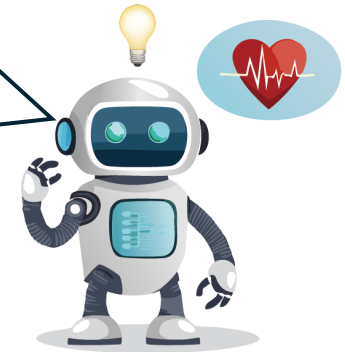
Bot : Five senses + Bio Info

+ ...

I can tell a condition
by **heart rate**!



How can I
match from
heart rate?



Let's change intonation to match!

Evaluating whether bots' **intonation** increases **human empathy**

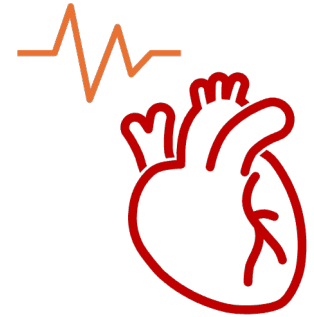
Why Focus on Heart Rate?

Unconscious & Honest Signal

Heart rate is controlled by the Autonomic Nervous System. Unlike facial expressions or voice tone, it is difficult to control or fake.

Internal State

Directly reflects the balance of stress (Sympathetic) vs. relaxation (Parasympathetic).



Robot's Capability

Wearable sensors allow robots to perceive this "invisible" information, enabling deeper understanding.

Related Works

Prior research has explored various ways to create empathetic agents:



Nishimura et al. (2021)

Demonstrated that
"Vocal Synchrony"
significantly boosts
positive affective
empathy.



Jonell (2019)

Used physiological
signals to adapt
conversational
agents to user state.



Saffaryazdi et al. (2025)

Utilized complex
emotion recognition
models using EEG
and heart rate to
drive digital humans.

However, does simple,
direct intonation feedback based on raw HR work?

Rationale for this Objective

Simplicity: Direct signal feedback is fast.

Robustness: HR signals are raw data.

Using them directly avoids "interpretation errors" by LLM.

Verification: We verify if **direct feedback** of internal states is sufficient for empathy.

Core Goal

To evaluate whether a bot's intonation, when directly adjusted to the user's heart rate, increases human empathy *without* complex emotion models.

Heart Rate Feedback : HRF

Algorithm

± 1 bpm Change
the heart rate

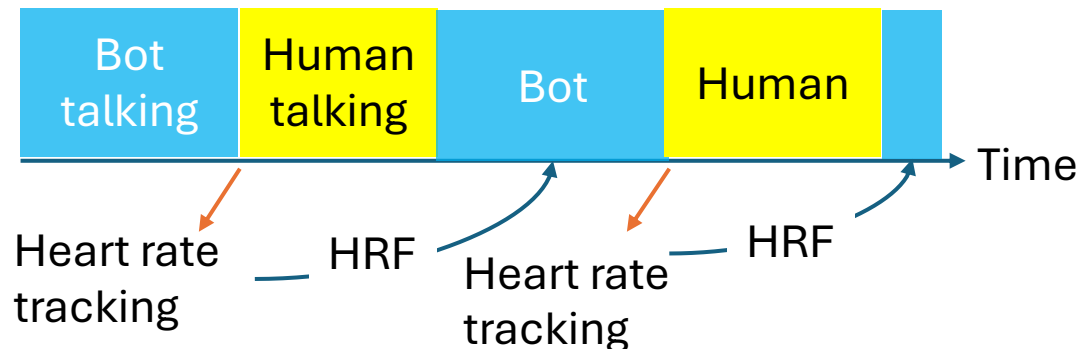


± 0.1 Change
the Intonation level

Reference: Resting heart rate
Inflection range: 0.0 to 2.0

We don't collect heart rates at all times, but only when a person starts speaking

- ➡ Mouth & vocal cord movement: **Minimal motion artifacts**
- ➡ Considering the time window of speech content impact
- ➡ Single-point HRF: Prevents **inconsistent intonation**



Physiological

Why? To measure *unconscious* biological reactions (stress/relax) that cannot be faked.

- ✓ **RMSSD:** Parasympathetic (Rest).
Higher = Calmer.
- ✓ **LF/HF Ratio:** Sympathetic
(Stress). Lower = Less Stress.

Questionnaire (VAS)

Why? To quantify the user's *conscious* psychological feeling of rapport and empathy.

- ✓ "Did the robot sympathize?"
(Empathy)
- ✓ "Did the robot listen?" (Listening)
- ✓ "How much stress?" (Stress)

The topic of "**Hobbies**" under three distinct conditions:



1. Fixed Intonation

The chatbot speaks with a constant, flat intonation (Level 1.0). This represents the standard, non-adaptive robotic response.



2. Proposed (HRF)

The chatbot's intonation actively changes based on the user's real-time heart rate deviations.



3. Sine Wave

The intonation changes mechanically in a sine wave pattern. This acts as a control for "random change."

Experimental procedure

1

Baseline

3 mins stable HR
measurement &
Practice.

2

Session 1

7-min Conversation

3

Session 2

7-min Conversation

4

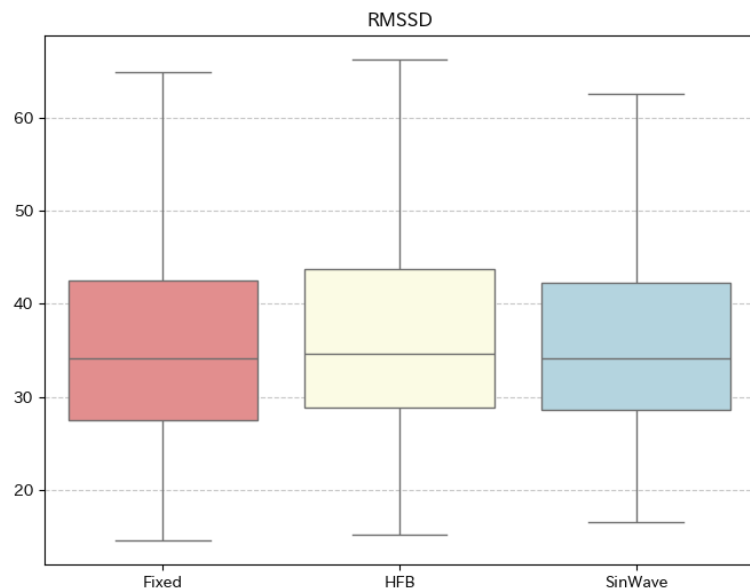
Session 3

7-min Conversation

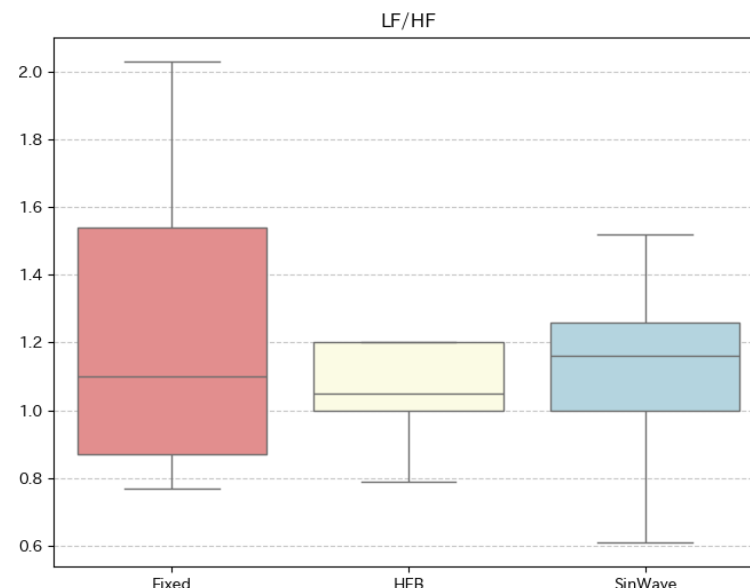
- The order of conversation was randomized by person.
- Questionnaires were conducted immediately after each session.

Result • Autonomic nervous

- **RMSSD** : Overall increase



- **LF/HF** : Overall decrease



= Fixed
 = HRF
 = SinWave

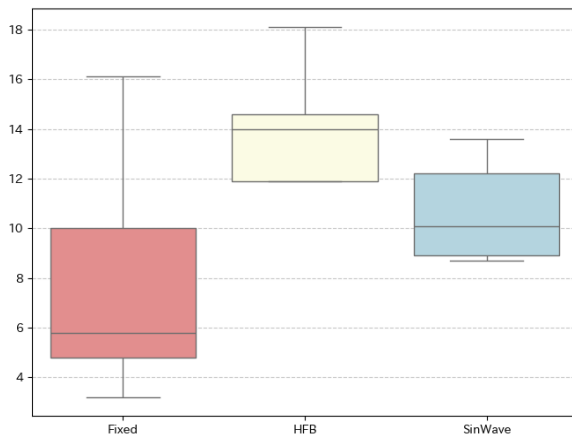
Although the changes were slight, the mean, median, and maximum values of RMSSD increased, while the median, maximum, and minimum values of LF/HF decreased.

➡ Decreases stress on the mind and body

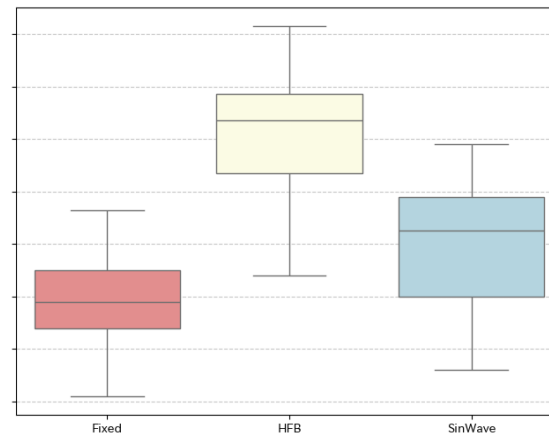
Result • Questionnaire

Evaluation of results: Changes have a big impact

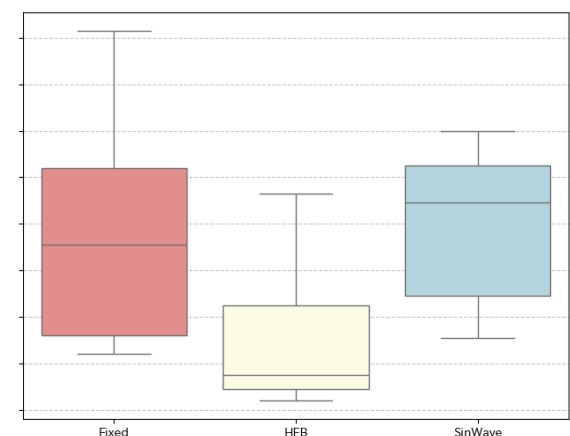
Empathy



Listening to me



Stress



■ = Fixed ■ = HRF ■ = SinWave

By changing the intonation, both the range and median stress levels changed compared to when the intonation was unchanged.

➡ The participants felt less stressed and were able to converse effectively.

Conclusions

- **Physiological Proof:** ANOVA confirmed significant main effect on RMSSD ($p < .001$).
The body **relaxes** when the robot adapts
- **Stress Reduction:** Both **physiological** (LF/HF) and **Questionnaire** metrics confirmed a reduction.

Significant HRF effects change in intonation caused by heart rate



Reduced mental and physical stress (biometric information)
Decreased stress and increased empathy (Questionnaire)



HRF affects people

Future works

- Improving system architecture to minimize **conversation delay**.
- Investigating **other parameters** (speed, pitch) driven by HRF.

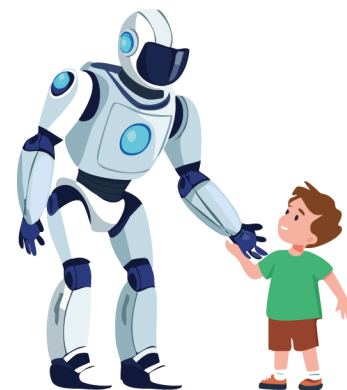
In the future,
it will be equipped with predictive coding of heart rate.

Social significance

- Conversations between robots and humans can foster greater empathy



Robots that blend into human society

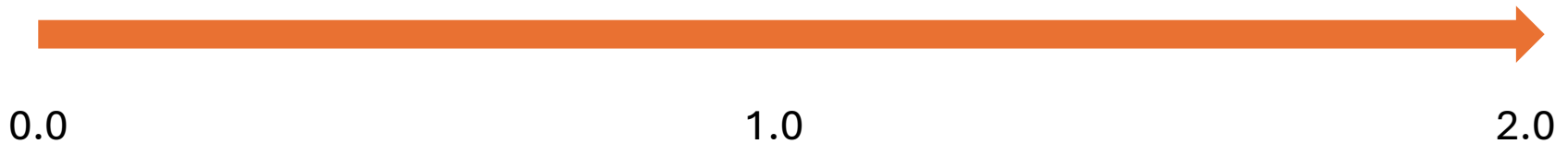


Thank you for listening

Supplementary Slides
are listed below.

Supplementary Slides 1

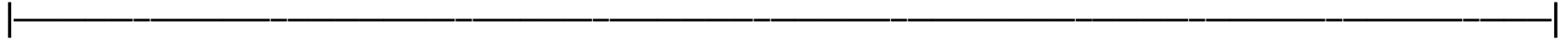
Intonation Levels



Supplementary Slides 2

Questionnaire(Visual Analog Scale : VAS)

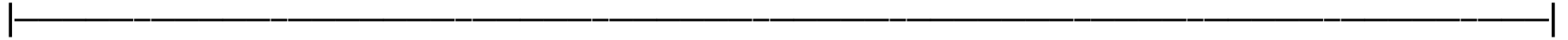
- ・ この会話中に感じたストレスの強さを教えてください。



全くストレスを感じない

非常にストレスを感じる

- ・ この会話では、相手に共感できましたか？



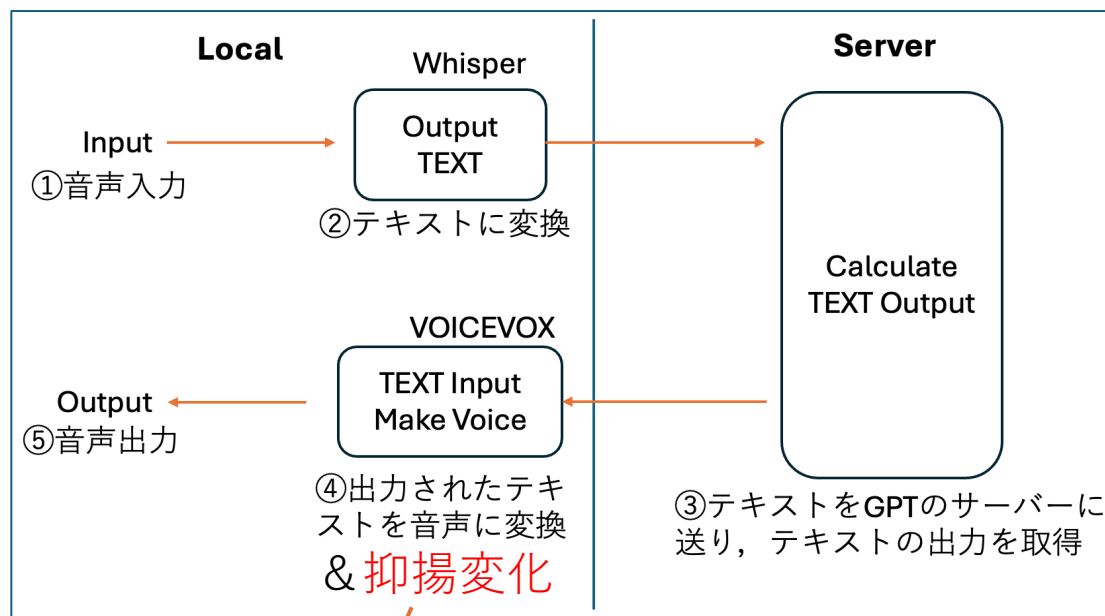
全くできなかった

非常によくできた

Supplementary Slides 3

会話システム

課題：リアルタイムで抑揚変化するシステムがない



心拍数フィードバック(HFB)とその影響

課題：どのように心拍数を反映するか？

抑揚がバラバラにならず，運動の影響を受けない ➡ **HFB**